

DATA ANALYTICS LEVEL III

Day 7: Data Joining & Transformation

Student Activities

Day 7 — Student Activities Overview

Unit of Competency: Prepare Data Sets (Core 2, wrap-up)

Learning Outcomes Assessed: LO1 Import, LO2 Clean, LO3 Filter (applied via JOINS and transformation)

Activity	Type	Duration	Weight
Core 2 Assessment: Data Preparation Project	Summative	2.5 hours	15% of final grade
Reading: Data Joining Concepts	Self-study	0.5 hours	N/A
Learning Journal Entry #7	Reflective	0.5 hours	Part of participation grade

Core 2 Assessment: Data Preparation Project

Type: Summative Assessment (15% of Final Grade)

Time Allowed: 2.5 hours

Files Needed: Day07_assessment_customers.csv, Day07_assessment_orders.csv, Day07_assessment_products.csv

Scenario

You are a data analyst at TechShop Philippines, an online electronics retailer. Your manager has given you three raw data files exported from different systems. They need a single, clean, analysis-ready dataset for the quarterly business review.

The three files are:

- Day07_assessment_customers.csv — Customer information (13 rows)
- Day07_assessment_orders.csv — Order transactions (19 rows)
- Day07_assessment_products.csv — Product catalog (6 rows)

Your task: Import, clean, join, and deliver a professional dataset.

Part A: Import & Clean (40 points)

Task 1: Import and Clean Customers (15 pts)

Import Day07_assessment_customers.csv into Excel (Power Query) or SQL. Fix ALL issues:

1. Remove duplicate rows (the same customer appears twice)
2. Trim leading/trailing spaces from names
3. Standardize city names to proper case
4. Standardize membership values to proper case (Gold, Silver, Bronze)
5. Handle missing email (document how you handled it)

Task 2: Import and Clean Orders (15 pts)

Import Day07_assessment_orders.csv. Fix ALL issues:

6. Remove the test/junk data row
7. Standardize date formats (all should be YYYY-MM-DD)
8. Handle the 'N/A' in the amount column
9. Verify all cust_id values exist in the customers file

Task 3: Import and Clean Products (10 pts)

Import Day07_assessment_products.csv. Fix:

10. Standardize category values to proper case

Part B: Join the Data (30 points)

Task 4: Create a Combined Dataset (20 pts)

Join all three cleaned tables into one combined dataset. The final dataset should include:

- Customer name and city (from customers)
- Product name and category (from products)
- Order date, quantity, and amount (from orders)

You may use either method:

- **SQL:** Use INNER JOIN or LEFT JOIN queries
- **Excel:** Use VLOOKUP/XLOOKUP or Power Query Merge

Task 5: Verify the Combined Dataset (10 pts)

After joining, verify:

11. How many rows are in the final combined dataset?
12. Are there any NULL values in key fields?
13. Are all customer names, product names, and categories correct?
14. Document your verification steps

Part C: Data Quality Report (30 points)

Task 6: Write a Data Quality Report (30 pts)

In a separate sheet or document, write a report that includes:

15. Summary of Issues Found (10 pts) — List every data quality issue you discovered across all 3 files. Be specific: how many duplicates, which fields had issues, what the problems were.
16. Cleaning Steps Taken (10 pts) — Describe each step you took to fix the issues. Include your SQL queries or Power Query steps.
17. Final Dataset Summary (10 pts) — How many rows/columns in the final dataset? How many unique customers? How many unique products? Total order amount? Brief summary of the data.

Grading Rubric (100 points)

Component	Points	Criteria
Customer cleaning	15	All issues found and fixed correctly
Order cleaning	15	Junk removed, dates fixed, N/A handled
Product cleaning	10	Category casing standardized
Combined dataset	20	Correct join, all fields present, no orphaned records
Verification	10	Row counts, NULL checks, documentation
Quality Report	30	Complete, specific, well-organized

Submit TWO files:

- Day07_Core2_Assessment_[YourName].xlsx — containing cleaned sheets + combined data + QA report
- Day07_Core2_SQL_[YourName].sql — your SQL queries (if you used SQL)

Learning Journal Entry #7

Time: 30 minutes

Reflect on your Core 2 journey. Write at least 5 sentences:

18. How confident do you feel about data preparation now compared to Day 4 when you first saw SQL?
19. What was the most challenging part of the Core 2 Assessment? How did you work through it?
20. If you had to explain JOINS to someone who only knows Excel, how would you describe it?
21. Which data preparation skill (filtering, cleaning, or joining) do you think is most important in a real job? Why?
22. What are you looking forward to in the next phase (data summarization)?

Save as: Day07_Journal_[YourName].docx

Summary of Submissions

Submission	Filename	Due
Core 2 Assessment	Day07_Core2_Assessment_[YourName].xlsx	Before next session
SQL Queries (if used)	Day07_Core2_SQL_[YourName].sql	Before next session
Learning Journal #7	Day07_Journal_[YourName].docx	Before next session